PHASE 3

Exploratory Data Analysis

A detailed analysis blueprint for Group 7's Gaza ACLED project

Purpose. Plan and document a question-led EDA using Phase 2 outputs. This guide specifies which tables and variables to use, what figures and tests to produce, and how findings should inform Phase 4.

At a glance

Guide item	Phase 3 direction
Primary aim	Describe patterns, relationships, and unusual periods in cleaned Gaza event records; refine the Phase 4 modeling plan.
Analysis inputs	Cleaned event master; Gaza-wide and district-week tables; post-October 7 land focal-incident table.
Minimum requirements	Univariate and bivariate/multivariate analysis; at least 8 purposeful visualizations; at least 2 hypothesis tests; written interpretations; 5-8 key insights and modeling plan.
Notebook sequence	Scope and data contract -> univariate -> bivariate -> grouped/time -> spatial -> hypothesis tests -> insights and Phase 4 plan.
Audience	Human-rights documentation researchers, civilian-protection analysts, investigative journalists, and research coordinators.

Decision supported

Prioritize locations, periods, and incident groups for source verification and contextual investigation.

Interpretation boundary: Describe records coded by ACLED and the project's reproducible focal-event rule. A targeting flag, association, cluster, or anomaly is not proof of intent, responsibility, causation, or legal classification.

This is a planning guide. Replace bracketed prompts with findings from the Phase 2 snapshot; do not report planned questions as results.

1. Analysis scope and data contract

Phase 2 narrowed the working analysis to Gaza and created distinct tables for event-level, temporal, district, and spatial questions. State the unit of analysis before making comparisons; an event, a week, and a coordinate are not interchangeable observations.

Table	Unit / period	Use in Phase 3	Restrictions / checks
acled_events_clean.csv	One Gaza event record; Jan 2022-Sep 2025	Before/after context; event, actor, type, fatality, and precision descriptions.	Keep all Gaza-coded records. Preserve optional actor blanks. Four records lack admin2.
acled_gaza_week.csv	One Gaza-wide week; Oct 7, 2023 onward	Weekly counts, shares, coverage checks, prior-week context.	Zero-event calendar weeks retained; final week may be partial; share undefined at zero denominator.
acled_district_week.csv	One district and week; Oct 7, 2023 onward	District comparisons, persistence, heatmaps, historical features.	Known-district records only. A zero recorded count is not evidence of no harm.
acled_focal_incidents.csv	One land-based focal incident; Oct 7, 2023 onward	Coordinate distribution and precision sensitivity before DBSCAN.	Coast-coded focal events excluded in this spatial file; repeated coordinates are expected.

Variable roles

Role	Variables	How to use / interpretation
Outcomes / denominators	`focal_event`, `political_violence`, `targeting_flagged`, `civilian_targeting`	Focal flag is derived from the recorded targeting flag and an exact Palestinian civilian actor label in either actor field. Use political-violence records as focal-share denominator.
Time	`event_date`, `period`, `month_start`, `week_start`, `week_end`, `complete_week`	Use period for before/after context; week/month for trends. Periods differ in duration: provide rates or shares with counts.
Categories	`event_type`, `sub_event_type`, `disorder_type`, `actor1`, `actor2`	Describe recorded categories and co-occurrence. Review notes before describing roles; actor2 blanks can be structural.
Severity	`fatalities`, `fatalities_log1p`, `high_fatalities_review`	Show raw counts and transformed display. Review flag is not a deletion rule.
Geography / quality	`admin1`, `admin2`, `location`, `latitude`, `longitude`, `geo_precision`, `time_precision`, `geo_precision_1`	Precision codes are categories. Map in geographic units; compare precision-1 subset as a sensitivity check.
Weekly features	`recorded_events`, `political_events`, `focal_events`, `focal_share`, `history_eligible`, `previous_week_focal`, `previous_four_weeks_focal`	Use for trends and history. Phase 4 predictors must use information available before the assessed week.

Scope rule: Keep all Gaza events for event-level and Gaza-wide summaries. Apply the district-missing restriction only to district results and coast exclusion only to land spatial analysis.

Sources: `DMA_Phase1.ipynb`, `DMA_PHASE2.ipynb`, and `[STUDENT] DS312_FINAL_PIT_GUIDE.pdf`, Phase 3.

2. Univariate and bivariate analysis

Start with distributions and group sizes. Compare variables that directly answer the business questions. Use a question-led title and a short interpretation for every selected figure.

Univariate checklist

Variable(s)	Recommended summary / plot	What to report
`fatalities`, `fatalities_log1p`	Histogram or ECDF; median, IQR, 90th/95th/99th percentiles; raw plus log1p view.	Describe zeros, center, spread, and tail. Retain original totals.
`event_type`, `sub_event_type`, `disorder_type`	Frequency table; sorted count/share bar chart.	Name dominant groups and state denominator; document multi-label categories.
`actor1`, `actor2`	Top-category frequencies overall and among focal events; missing counts.	Explain structural actor2 blanks. Labels alone do not establish roles.
`targeting_flagged`, `political_violence`, `focal_event`	Counts and proportions, optionally split by period.	Keep ACLED targeting flag distinct from project focal-event definition.
`geo_precision`, `time_precision`	Frequency counts and shares by code.	Treat as quality categories, not continuous measurements.
`admin2`, `period`, `week_start`	Frequency and activity summaries at matching area/time unit.	State missing-district handling; distinguish totals from rates/shares.

Bivariate priorities

Question / pairing	Recommended display	Guardrail
Did focal share differ before/after Oct 7? `period` x focal among political violence	Numerator/denominator table plus rate plot.	Show counts and share; periods differ in length and pre-period focal group is sparse.
Which event types have higher focal prevalence? `event_type` x focal	Within-type count/share bars and cross-tab.	Raw counts reflect type volume; show within-type denominator.
Which actor labels recur with event types among focal events? actor x type	Top-label cross-tab or heatmap; review event notes for selected cases.	Do not label actor role without supporting coding/context.
How do fatalities vary by focal status/type? fatalities x group	Boxplot plus ECDF or median/IQR table.	Skew and zeros make mean-only summaries misleading; show sample sizes.
Does data precision relate to apparent pattern? precision x focal	Cross-tab and precision-stratified counts/maps.	Differences can reflect documentation quality or coverage.
Which districts have larger count or share? admin2 x weekly counts	District table and week-aligned chart.	Counts and shares answer different questions; display denominator.

Chart interpretation prompt: What is visible? How large or stable is the pattern? Which denominator, missingness, precision, or recording limitation affects how strongly it can be stated?

3. Multivariate, temporal, and spatial analysis

Use grouped analysis to reveal a meaningful interaction or persistence pattern. Add dimensions only when they help answer a business question.

Analysis	Variables / unit	Recommended approach	Phase 4 relevance
District x week counts	`admin2`, `week_start`, `focal_events`	District-week heatmap with shared week axis; small-multiple lines for selected districts.	Concentration and persistence by district.
District x week share	`focal_events`, `political_events`, `focal_share`	Plot only with denominator > 0; show denominator alongside.	Relative focal activity versus overall political violence.
Event type x period x focal	`event_type`, `period`, `focal_event`	Faceted rate chart or compact table.	Whether event composition differs between descriptive periods.
Actor x event type	`actor1` / `actor2`, type among focal events	Cross-tab or heatmap for common documented labels.	Identify recurring coded combinations for note review.
Historical activity	`previous_week_focal`, `previous_four_weeks_focal`, `history_eligible`	Compare distributions and coverage; summarize eligible history by district.	Judge whether lag features are interpretable and sufficiently populated.
Spatial distribution	coordinates, precision, period/date	Map/scatter and density view; compare full and precision-1 subsets.	Assess DBSCAN relevance and precision stability.

Temporal and spatial rules

- Use the Phase 2 incident file only for land-based focal spatial analysis; do not apply its coast exclusion to all Gaza summaries.
- For DBSCAN, use geographic coordinates with haversine distance and radians; do not standardize latitude and longitude independently.
- Repeated coordinate pairs can mean multiple events share a named location; describe recorded concentrations, not exact attack points.
- Mark the incomplete final week. Keep zero-event weeks and distinguish zero recorded activity from evidence of no harm.
- Lag variables describe prior history. Do not use same-week focal counts or shares as predictors of that same week's outcome.

Visualization backlog - choose at least 8

ID	Question and fields	Suggested chart
V1	How are fatalities distributed?	Raw histogram/ECDF plus log1p display.
V2	Which event types dominate?	Sorted count/share bar chart.
V3	Which actor labels recur among focal events?	Top-label bar chart and missingness note.
V4	Did focal share change by period?	Period numerator/denominator and rate plot.
V5	How does focal prevalence vary by event type?	Within-type rate bars with counts.
V6	How do fatalities vary by group?	Boxplot or ECDF with sample sizes.
V7	Which districts have high focal weeks?	District-week heatmap.
V8	How does weekly activity change?	Two aligned count and share time series.
V9	Where are land focal incidents recorded?	Map/scatter or density view.
V10	Does precision restriction alter the map?	Matched full vs precision-1 maps.

Each chart needs a title that states the question, labeled axes and legend, and a 1-2 sentence interpretation beneath it.

4. Hypothesis testing specification

The course guide asks for at least two hypotheses based on the business questions. State null and alternative hypotheses before reporting results. Include test, p-value, assumption checks, effect size, and a plain-language conclusion. Tests are exploratory and do not establish causation.

Hypothesis A - period and focal status

Item	Specification
Question	Did the proportion of political-violence event records classified as focal differ before versus after Oct 7, 2023?
H0 / H1	H0: period and focal status are independent among political-violence records. H1: they are associated.
Table	Rows: period. Columns: focal yes/no. Restrict to `political_violence == True`.
Candidate test	Pearson chi-square if expected frequencies are adequate; otherwise a justified exact or Monte Carlo procedure.
Effect size	Cramer's V, alongside period-specific numerator, denominator, and share.
Caution	Unequal period lengths, sparse earlier focal count, and event dependence across time/place limit a simple independence interpretation. Show counts and weekly context.

Hypothesis B - event type and focal status

Item	Specification
Question	Does focal-event prevalence differ across recorded event types?
H0 / H1	H0: event type and focal status are independent among political-violence records. H1: at least one type has a different prevalence.
Table	Rows: `event_type`. Columns: focal yes/no. Restrict to political-violence records.
Candidate test	Pearson chi-square if expected counts are adequate; otherwise Monte Carlo resampling or another justified method.
Effect size	Cramer's V, plus within-type proportions and category sample sizes.
Caution	This is an association in this export, not evidence that event type causes focal classification.

Results table to complete

Hypothesis	Test / checks	Statistic	p-value	Effect size	Conclusion / limitation
A: period x focal	[Fill]	[Fill]	[Fill]	[Fill]	[Fill]
B: event type x focal	[Fill]	[Fill]	[Fill]	[Fill]	[Fill]

Test preflight

- Confirm the table population and denominator match the stated hypothesis; report included/excluded counts.
- Inspect expected cell counts. Do not blindly use an asymptotic p-value when cells are sparse; document a suitable alternative and its settings/seed.
- Choose and report a significance threshold, such as alpha = 0.05. Label tests exploratory and disclose additional comparisons.
- Discuss related events within weeks or locations; a p-value does not remove dependence or reporting bias.

5. Reusable functions and notebook design

Reusable EDA functions are helpful for consistency but are not a Phase 3 requirement. In the course guide, a reusable cleaning pipeline appears in the advanced Phase 2 criteria. Phase 3 requires the analyses, purposeful charts, interpretations, tests, and insights.

Recommendation: Use a few small plotting helpers for repeated chart patterns. Keep question-specific filters, denominators, test choices, and interpretation visible in the notebook.

Suggested helper boundaries

Helper	Good responsibility	Keep explicit in analysis cells
`plot_category_counts(data, column, title, order)`	Consistent count/share styling and labels.	Which categories are grouped; count versus share meaning; interpretation.
`plot_numeric_distribution(data, column, title, bins)`	Reusable histogram/ECDF framing.	Raw vs log scale, robust summaries, what skew means.
`summarize_group_rate(data, group, numerator, denominator)`	Return numerator, denominator, rate; handle zero denominator transparently.	Why this denominator answers the question.
`plot_weekly_series(data, date, value, title)`	Consistent date-axis style and visual labels.	Count or share meaning, incomplete-week handling, eligible history.
Hypothesis test helper	Usually keep crosstab and assumptions visible.	Population filter, test selection, expected counts, resampling settings, effect size.

Suggested notebook outline

Section	Contents
3.1 Purpose and scope	Audience, decision, Gaza scope, units, interpretation boundary.
3.2 Data contract	Load Phase 2 tables; restore dates/categories; show shapes and completeness.
3.3 Univariate	Counts, categories, fatality distribution, flags, actors, precision.
3.4 Bivariate	Period/type comparisons, actor/event profiles, severity and precision comparisons.
3.5 Grouped, temporal, spatial	District-week, weekly trends, historical features, maps and precision sensitivity.
3.6 Hypotheses	Two written hypotheses, test diagnostics, effect sizes and conclusions.
3.7 Insights and Phase 4 plan	5-8 evidence-backed insights; selected methods, inputs, baselines, sensitivity.
3.8 References and reproducibility	Snapshot, filters, sources, notebook execution checklist.

Interpretation template for each figure

Question: What does this figure answer? Result: What pattern and counts/rates are visible? Meaning: Why does it matter? Limit: What recording, denominator, missingness, or precision caveat applies?

Keep interpretations specific and concise. Do not repeat the title or turn an association into a causal claim.

6. Acceptance checklist and Phase 4 handoff

Use this as the final review gate. Each checked item should be supported by a chart, table, test report, or written explanation.

Analysis and evidence

- At least 8 purposeful charts; each has a question-led title, labeled axes/legend, and 1-2 sentence interpretation.
- Univariate, bivariate, and multivariate/grouped analyses are visibly organized; temporal and spatial views answer planned questions.
- At least two hypotheses include test name, p-value, expected-count/assumption checks, effect size where appropriate, and plain-language conclusion.
- Focal shares use matching political-violence denominators; counts, rates, and shares are clearly distinguished.
- District-week comparisons use consistent weeks and explain zero-denominator or missing-district cases.
- Five to eight key insights connect evidence to the documentation decision and proposed Phase 4 work.

Data and interpretation controls

- Event-level, weekly, and spatial units are not mixed without an explicit aggregation step.
- Optional actor blanks remain unfilled; precision codes are categories; high-fatality flags do not delete events.
- Four district-missing events remain in Gaza-wide summaries and are excluded only from district totals.
- Coast exclusion applies only to land spatial analysis; incomplete weeks are marked.
- Zero recorded events are not described as proof that no harm occurred.
- No result is presented as proof of intent, responsibility, causation, or legal classification.

Phase 4 handoff

Technique	EDA evidence to cite	Inputs	Baseline / sensitivity
DBSCAN spatial clustering	[Concentration and precision stability]	Land focal coordinates in radians; haversine distance.	Compare with district counts; vary radius/min_samples and precision-1 subset.
Weekly anomaly detection	[Spikes, persistence, eligible history]	District-week focal counts/shares and lagged history features.	Trailing median/MAD baseline; compare settings or Isolation Forest; chronological evaluation.

Carry-forward limitations

- ACLED may underreport events, and reporting coverage may vary across place and time.
- Fatalities are reported event totals and do not automatically identify civilian deaths.
- Coordinates and dates have varying precision; repeated coordinates may reflect shared named locations.
- Events may be related across geography and time, affecting simple independence assumptions.
- Findings describe a historical snapshot, not current conditions.

References and provenance

1. Group 7, DMA_Phase1.ipynb: Phase 1 business questions, candidate variables, and preliminary data-mining plan.
2. Group 7, DMA_PHASE2.ipynb: processing decisions and Phase 2 analysis tables.
3. DS312, [STUDENT] DS312_FINAL_PIT_GUIDE.pdf: Phase 3 EDA requirements and checklist.
4. ACLED Codebook and export documentation, as cited in the Phase 1 and Phase 2 notebooks.

Prepared as a planning guide for Group 7 Phase 3. Replace bracketed prompts with findings from the project snapshot before submission.